STAFF REVIEW · H446 211 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Index 3 is outside indexes 0–2; a C# IndexOutOfRangeException occurs.

2. O(n), as it may inspect every element.

3. C, because a stack is LIFO.

4. Bucket 2.

I Do — source worked steps

1. Predict first — Start with [2,7]. The top of the stack is 7.

2. Check the method — Pop removes 7, leaving [2].

3. Check the method — Push 5 gives [2,5]; the top is now 5.

We Do — guided CHECK

1. Check the method
Dequeue removes X, leaving [Y].
Enqueue Z produces [Y,Z]. The next dequeue removes Y.

You Do — source mark schemes

1. [3] First Pop removes 2 [1]; second Pop removes 9 [1]; remaining bottom-to-top [4] [1].

2. [4] Initial front-to-back [5,8,3] [1]; dequeue 5 [1]; enqueue 2 gives [8,3,2] [1]; next removal is 8 because FIFO [1].

3. [6] Root 40 [1]; 20 left of 40 [1]; 60 right of 40 [1]; 25 is the left child of 30 [1]. Increasingly sorted input can produce a chain [1], with search up to O(n) [1].

Exit Ticket — source answer

1. False. It removes the oldest enqueued item.